

Active GLiNER: Which Examples Should LLMs Label?

Abhishek Thomas

Independent researcher, Paris, France
abhi@recolli.com

Abstract

Small extraction models such as GLiNER can be trained on labels produced by a large language model (LLM), which makes annotation cheap but noisy. We ask a simple question: when the LLM can label only a few hundred examples, should the small model choose which ones? We study one round of uncertainty-based selection with a GLiNER2.5 student and LLM teachers on four extraction tasks and six datasets: named entity recognition (CleanCoNLL, BC5CDR, MIT Movie), relation prediction (CrossRE), multi-label classification (Hallmarks of Cancer) and slot extraction to JSON (MASSIVE, English and French). Our pre-registered primary contrast is the interaction between selection and label source: whether choosing the student’s least confident examples helps less when an LLM, rather than ground truth, provides the labels. The answer depends on the task. For NER, min selection loses to random under both label sources (min minus random is -6.96 F1 points with LLM labels), mostly because it picks sentences where the student predicts nothing; ranking those last reverses the loss on CleanCoNLL. For relations, min helps with ground truth, and LLM label errors remove most of that gain. LLM errors concentrate where the student is unsure only for NER. For NER, a student trained on 400 LLM-labelled sentences comes close to its teacher (75.28 against 76.36 F1) at 165 times the speed, and on some datasets students beat their teacher. Replicating an earlier thesis on all datasets, we find that its three uncertainty scores (mean confidence, MSE, MNLP) select identical sentences, because every pool holds more empty-prediction sentences than the budget, and that half ground truth matches full ground truth only on MIT Movie. We release the code as a command-line tool, together with all teacher labels, adapters and run logs.

1 Introduction

Information extraction models that run on a CPU in milliseconds are attractive for production, but they need labelled data for every new domain. Large language models can now produce those labels (Zhang et al., 2023; Kholodna et al., 2024), and a small student trained on LLM labels can match or beat its teacher at a fraction of the cost (Vedula et al., 2024). GLiNER (Zaratiana et al., 2023) and GLiNER2 (Zaratiana et al., 2025) made such students general: one encoder handles entities, relations, classification and structured output from a schema.

If an LLM labels the data, the next question is which data it should label. Active learning answers this for human annotators: pick the examples the current model is least sure about. Yet active learning often fails to beat random selection (Ghose and Nguyen, 2024; Yu and Fathima, 2026; Mittal et al., 2019), and LLM labels change the picture. The examples a student finds hardest may also be the examples an LLM gets wrong, so uncertainty could select exactly the noisiest labels. Hakimi et al. (2026) even find that labelling the whole pool with an LLM beats active learning.

We test this directly. Our question is: *when does student uncertainty improve one-round LLM annotation, and when does teacher error erase that benefit?* We make four contributions.

1. A pre-registered comparison of uncertainty and random selection, crossed with ground truth and LLM labels, on four tasks and six datasets, with five seeds on primary cells, paired bootstrap intervals and Holm correction.
2. An analysis of teacher errors by student confidence: do LLM mistakes concentrate where the student is unsure (Q2), do different LLMs

fail on the same examples (Q3), and do larger teachers label better (Q4)?

3. Cost, latency and size views of students and teachers, including a Pareto front and a quality-latency quadrant chart.
4. An open command-line tool that runs the whole loop (score, select, label with any OpenAI-compatible endpoint, train with LoRA, report), with every paper run reproducible through it.

This paper builds on the author’s master thesis, which studied the same loop for NER on MIT Movie with GLiNER and Gemma 3. Section 5 lists what the thesis did, what it found, and which of its protocol problems this study fixes.

2 Related work

Active learning with LLM annotators. LL-MaAA (Zhang et al., 2023) uses an LLM as the annotator inside active learning for NER and relation extraction. Kholodna et al. (2024) report that GPT-4 labels in the loop are 42 times cheaper for low-resource NER. ALLabel (Chen et al., 2025) selects demonstrations for the LLM rather than training data for a student. Qi et al. (2026) combine several small LLMs and use their disagreement to flag errors, and HyPAC (Zeng et al., 2026) routes examples between fast and slow LLMs and humans with an error guarantee. Active knowledge distillation (Luccioli et al., 2025) sends uncertain examples to an LLM teacher for classification. We differ in the student (a schema-driven GLiNER2.5 model), the breadth of tasks, and the explicit test of whether teacher error cancels the gain from selection.

Doubts about active learning. Random selection is a strong baseline (Ghose and Nguyen, 2024; Mittal et al., 2019). For NER, Yu and Fathima (2026) find uncertainty and diversity rules unstable, and a scoping review finds that only 13 of 62 studies report run time (Kohl et al., 2024). Practitioners cite setup effort and unclear savings as blockers (Romberg et al., 2025). We therefore pre-register the contrast, report seeds and intervals, and release a tool.

LLM-to-small-model distillation. GLiNER-BioMed (Yazdani et al., 2025) and FiNERweb (Golde et al., 2025) train GLiNER-style students

on LLM labels without selection. Vedula et al. (2024) show that a distilled BERT can beat its teacher. Merdjanovska et al. (2026) reweight noisy LLM NER labels by error type.

GLiNER and pre-annotation. The closest work to ours outside arXiv uses zero-shot GLiNER to pre-annotate hiking descriptions for human correction and retrains over three rounds (Tafer et al., 2026); it has no selection rule and no random comparison.

Uncertainty in extraction. Span probabilities can stay informative under domain shift (Zheng et al., 2026); calibration metrics for NER are discussed by Hashimoto et al. (2024). Classic active learning for NER uses sequence-level uncertainty such as MNLP (Shen et al., 2017).

3 Setup

3.1 Tasks and data

Table 1 lists the six datasets. We chose sets that are small (at most about 15k training sentences), carefully annotated, and cover four output structures.

For CrossRE we use an adapted protocol: entity mentions are given as input to every model, and the task is to predict all relation types for every ordered pair of mentions, including pairs with no relation. End-to-end triple extraction was too hard for both students and teachers to support a comparison (pilot micro-F1 below 0.08).

3.2 Student

The student is GLiNER2.5-multi (Fastino, 2026) (287M parameters, mDeBERTa-v3-base encoder), loaded from a pinned snapshot, trained with LoRA (Hu et al., 2021) through the gliner2 trainer. Every run starts from a fresh copy of the base model. Hyperparameters come from one Optuna search per task on the dev set (20 trials, the first being the vendor-recommended recipe), frozen before any main run. Training stops after three dev checks without improvement, and the checkpoint with the best dev micro-F1 is kept. CrossRE is the exception: its dev score moves up and down between checks, so every CrossRE run trains all 1000 steps and keeps its best dev checkpoint. For CrossRE the student scores each ordered mention pair with the checkpoint’s pretrained relation scorer, applied to the given mention offsets, so that its zero-shot confidence is meaningful for selection. All runs

Task	Dataset	Pool	Dev	Test	Labels
NER	CleanCoNLL (Rücker and Akbik, 2023; Tjong Kim Sang and De Meulder, 2003)	13,957	3,233	3,427	4 types
NER	BC5CDR (Li et al., 2016)	5,228	5,330	5,865	chemical, disease
NER	MIT Movie, corrected (Liu et al., 2013; Galileo, 2022)	8,797	978	2,443	12 types
Relations	CrossRE, 6 domains (Bassignana and Plank, 2022)	2,519	300	2,446	17 relation types
Multi-label	Hallmarks of Cancer (Baker et al., 2016)	12,119	1,798	3,547	10 hallmarks
Slots (JSON)	MASSIVE 1.1 en-US and fr-FR (FitzGerald et al., 2022)	11,514	2,033	2,974	55 slot types

Table 1: Datasets. MIT Movie has no official dev split: we hold out a seeded 10% of train. CrossRE pools train and dev and holds out 300 sentences as dev. Documents never cross splits in BC5CDR, CleanCoNLL and Hallmarks. Sentences with no gold label stay in the pool.

use deterministic GPU kernels in full fp32 precision without TF32, so a seed fixes the result.

3.3 Teachers

The main teacher is Gemma 4 12B (QAT, 4-bit GGUF) served locally with llama.cpp at temperature 0 with JSON-schema constrained output. One prompt per task gives the label names and, where a dev check showed it helped, one short definition per label from the annotation guidelines. The same prompt is used for every teacher. Teacher answers are validated: labels must be allowed and spans must occur exactly in the text, starting and ending on a word boundary of the student’s tokeniser. A mention that occurs only inside a longer word (for example the rating “r” inside “rated”) is dropped and counted as a teacher error. Teacher labels are cached, so every arm and seed sees the same labels for the same sentence.

3.4 Selection

From a frozen pool, a selector picks N sentences in one round. **Random** samples uniformly. **Min** scores every sentence with the zero-shot student and picks the lowest sentence confidence: the minimum span confidence for NER (no prediction counts as 0), the minimum distance of any label probability to the threshold for classification and relations, and for slots the minimum value confidence or, if nothing is predicted, one minus the highest below-threshold candidate. **Diversity** (NER only) clusters student sentence embeddings with k -means and takes the sentence nearest each centroid. Pool scores and embeddings are computed once and shared, so paired arms select identical sentences. Because “no prediction counts as 0” makes min prefer sentences where the student predicts nothing, a pre-registered sensitivity block on CleanCoNLL ranks such sentences last instead.

3.5 Protocol and statistics

Budgets are $N \in \{100, 400\}$ sentences (CrossRE $\{100, 200\}$). Primary cells (min and random, ground truth and Gemma, at the larger budget) have five seeds, all other cells three. Each seed sets pool tie-breaks, data order and LoRA initialisation. We report strict micro-F1 on the official test set; the test set is used once, for final scores only. Two references bound every cell: a lower bound from four zero-shot GLiNER2 models (small, base and multi from GLiNER2.5, and GLiNER2 large), each with its threshold chosen on dev, and an upper bound from training the student on the whole pool, once with ground truth and once with Gemma labels.

The pre-registered primary contrast, per task, is the interaction

$$\Delta = (\text{min} - \text{random})_{\text{Gemma}} - (\text{min} - \text{random})_{\text{ground truth}},$$

paired by seed and, for NER, pooled with equal weight over the three datasets. The four interactions form one Holm family; the four practical contrasts $(\text{min} - \text{random})_{\text{Gemma}}$ form a second. We report seed spread, paired test-set bootstrap intervals (Dror et al., 2018) and ASO (Dror et al., 2019; Ulmer et al., 2022).

4 Results

All numbers below come from 720 finished runs through active-gliner analyse. Intervals are 95% paired bootstrap intervals over test sentences; they do not include seed variation, so we give the seed standard deviation (SD) of each contrast as well.

Q1: does uncertainty selection help, with ground truth and with LLM labels? Table 2 gives every cell and Figure 1 the primary contrast. The interaction Δ is 3.32 F1 points for NER (95% interval 2.89 to 3.80, Holm-adjusted

Dataset	Labels	Zero-shot	Random	Min	Min, empty last	Diversity	Whole pool
CleanCoNLL	Ground truth	55.7	85.7 $\pm$ 0.9 (5)	70.7 $\pm$ 4.1 (5)	84.5 $\pm$ 0.6 (3)	86.3 $\pm$ 0.9 (3)	89.6
	Gemma 4 12B		72.2 $\pm$ 2.0 (5)	65.2 $\pm$ 1.7 (5)	75.1 $\pm$ 0.3 (3)	70.6 $\pm$ 0.4 (3)	74.0
BC5CDR	Ground truth	64.7	82.5 $\pm$ 0.5 (5)	72.4 $\pm$ 2.2 (5)	–	82.3 $\pm$ 0.5 (3)	87.4
	Gemma 4 12B		75.9 $\pm$ 0.7 (5)	65.4 $\pm$ 2.6 (5)	–	75.7 $\pm$ 1.6 (3)	79.5
MIT Movie	Ground truth	55.5	84.2 $\pm$ 0.2 (5)	78.5 $\pm$ 0.7 (5)	–	84.7 $\pm$ 0.7 (3)	88.4
	Gemma 4 12B		77.7 $\pm$ 1.7 (5)	74.4 $\pm$ 1.3 (5)	–	78.0 $\pm$ 0.3 (3)	79.7
CrossRE	Ground truth	8.4	33.2 $\pm$ 2.2 (5)	37.8 $\pm$ 0.4 (5)	–	–	41.0
	Gemma 4 12B		15.5 $\pm$ 1.4 (5)	16.9 $\pm$ 0.4 (5)	–	–	12.5
Hallmarks	Ground truth	13.2	43.0 $\pm$ 3.3 (5)	40.6 $\pm$ 1.3 (5)	–	–	42.3
	Gemma 4 12B		38.9 $\pm$ 1.5 (5)	35.6 $\pm$ 0.8 (5)	–	–	40.5
MASSIVE en	Ground truth	31.6	70.0 $\pm$ 1.1 (5)	69.7 $\pm$ 0.7 (5)	–	–	75.4
	Gemma 4 12B		48.5 $\pm$ 1.5 (5)	48.8 $\pm$ 1.3 (5)	–	–	54.1
MASSIVE fr	Ground truth	26.2	62.0 $\pm$ 1.0 (3)	64.5 $\pm$ 0.9 (3)	–	–	–
	Gemma 4 12B		46.7 $\pm$ 0.7 (3)	47.9 $\pm$ 0.4 (3)	–	–	–

Table 2: Test micro-F1 (%) at the main budget ($N = 400$; CrossRE $N = 200$): mean $\pm$ seed SD (number of seeds). Zero-shot is the untrained student with a dev-chosen threshold. Whole pool is one seed. Generated by active-gliner analyse.

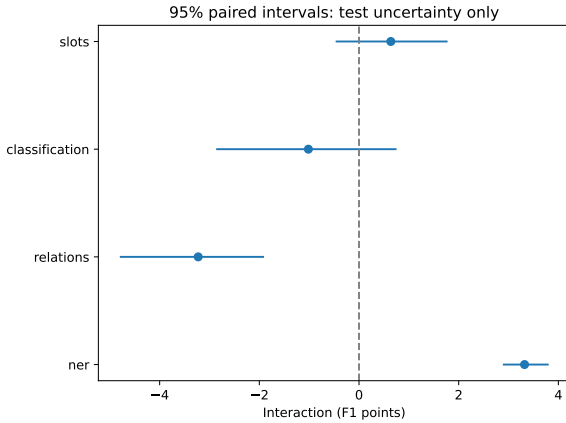


Figure 1: Primary interaction Δ per task with 95% paired bootstrap intervals (test uncertainty only). Positive: min loses less against random with Gemma labels than with ground truth.

$p < 0.001$, seed SD 2.93), -3.23 for relations (-4.80 to -1.91 , $p < 0.001$, SD 2.97), -1.02 for classification (-2.87 to 0.75 , $p = 0.512$) and 0.64 for slots (-0.47 to 1.78 , $p = 0.512$). The practical contrast, min minus random with Gemma labels, is -6.96 for NER (-7.34 to -6.63 , $p < 0.001$), 1.37 for relations (0.61 to 2.24 , $p < 0.001$), -3.37 for classification (-4.48 to -2.32 , $p < 0.001$) and 0.36 for slots (-0.37 to 1.10 , $p = 0.390$).

The two contrasts must be read together. For NER, the interaction is positive because min loses to random under both label sources and loses less with Gemma labels; there is no gain for teacher error to erase. For relations, the pre-registered pattern appears: with ground truth, min beats random (37.80 against 33.20), and with Gemma labels most of that gain is gone (16.88 against 15.51). For classification, min loses under both sources.

For slots, the selectors differ little, and the practical interval includes zero. The NER interaction varies across datasets: its seed SD is close to its mean, and on BC5CDR min loses about as much under both label sources, so the interaction there is near zero (Table 2).

On CleanCoNLL, the no-prediction sensitivity block separates two effects. With ground truth, min scores 70.67, min with empty sentences ranked last 84.46, and random 85.66. With Gemma labels the same three arms score 65.20, 75.07 and 72.24, and training on the whole Gemma-labelled pool scores 73.98. Most of the loss of min therefore comes from the rule that sentences with no prediction count as least confident: min then spends the budget on sentences where the student predicts nothing. With that rule reversed and Gemma labels, 400 selected sentences match or exceed the whole labelled pool on this dataset. This is one dataset and a sensitivity analysis, not a pre-registered contrast.

Q2: are LLM errors where the student is unsure? For NER, yes (Figure 2). Gemma’s F1 rises from the student’s least sure quarter to its most sure quarter: 70.32 to 87.98 on BC5CDR, 69.24 to 79.94 on MIT Movie and 76.23 to 81.39 on CleanCoNLL. So uncertainty selection sends the teacher the sentences it labels worst. For the other tasks the pattern reverses at the top: the teacher is weakest where the student is most sure (CrossRE 29.41 against 12.72, Hallmarks 45.76 against 39.72, MASSIVE English 49.57 against 44.54). The bins alone do not predict the sign of the interaction: NER, where teacher errors follow student doubt, has a positive Δ , and relations,

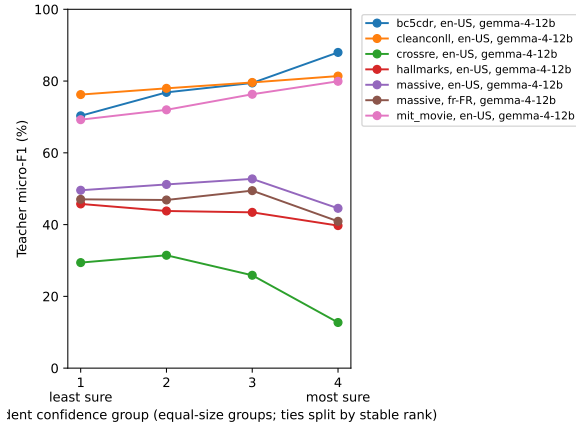


Figure 2: Gemma 4 12B micro-F1 on the pool, by the zero-shot student’s sentence confidence (four equal-size groups).

where they do not, has a negative one. The level of teacher error matters more: Gemma reaches only 22.62 F1 on CrossRE test, so any selected set carries many wrong labels.

Q3 and Q4: do teachers fail on the same examples, and do larger teachers label better? We compare Gemma 4 12B with the smaller Gemma 4 E4B on frozen, stratified test samples (up to 1,000 sentences, whole documents where the corpus has them), weighting each sentence by its inclusion probability. The larger teacher is better on five of six English datasets: by 6.59 points on BC5CDR (4.15 to 9.10), 8.93 on CleanCoNLL (4.24 to 13.92), 12.85 on MIT Movie (10.45 to 15.26), 4.31 on Hallmarks (0.15 to 8.47) and 13.61 on MASSIVE (10.86 to 16.40). On CrossRE both fail alike (22.39 against 21.82; gap interval -1.85 to 2.97). The two teachers err on largely the same sentences: the Jaccard overlap of their error sentences is between 0.60 (MIT Movie) and 0.70 (CleanCoNLL), and 0.90 on CrossRE. Both teachers come from one model family; the planned teachers from other families were not run (Limitations), so we cannot say whether errors are shared across families. The CleanCoNLL sample holds few documents, so its level estimate (65.16 for 12B) differs from the full test score (76.09); the paired gap uses the same sentences for both teachers and is not affected by this.

Q5 and Q6: better teachers, better students? Can the student beat its teacher? A better teacher gives a better student where we can test it: with 400 random sentences, E4B labels train a student to 68.23 on CleanCoNLL and 35.82 on Hall-

Dataset	Ground-truth share					Routed
	0%	25%	50%	75%	100%	
CleanCoNLL	65.20	65.18	67.38	68.11	70.67	62.86
BC5CDR	65.36	64.87	69.70	71.92	72.37	59.49
MIT Movie	74.38	75.03	78.16	79.06	78.53	75.70
CrossRE	16.88	21.00	27.61	32.63	37.80	–
Hallmarks	35.58	35.35	37.25	38.39	40.62	–
MASSIVE en	48.83	53.73	58.43	64.21	69.71	–
MASSIVE fr	47.89	54.02	56.50	61.20	64.47	–

Table 3: Mixing: test micro-F1 (%) with min selection at the main budget, mean of three seeds (0% and 100%: five seeds; MASSIVE fr: three). The rest of each set keeps Gemma 4 12B labels. Routed: the ground-truth share goes to the least confident sentences.

marks, against 72.24 and 38.95 with 12B labels (three seeds each). The student can beat its teacher. Trained on the whole Gemma-labelled pool (one seed), it exceeds Gemma’s own test F1 on MIT Movie (79.69 against 73.95), BC5CDR (79.54 against 79.04) and MASSIVE English (54.09 against 48.75); with only 400 random sentences it already does so on MIT Movie (77.68) and MASSIVE French (46.74 against 45.56). On CrossRE it stays far below the teacher (12.54 against 22.62), and the whole-pool CrossRE student trained on Gemma labels scores lower than the one trained on 200 random sentences.

Mixing ground truth into LLM labels. The thesis found on MIT Movie that a share of ground truth closes most of the gap to a fully human-labelled set. We repeat this on every dataset with min selection at the main budget: a random fraction of the selected sentences gets ground truth and the rest keeps Gemma labels (three seeds per cell; Table 3). MIT Movie reproduces the thesis: 50% ground truth (78.16) already matches all ground truth (78.53). No other dataset does. On CrossRE and MASSIVE every extra share helps by a similar step (CrossRE 16.88, 21.00, 27.61, 32.63, 37.80), and on CleanCoNLL, BC5CDR and Hallmarks the first 25% gives almost nothing. So a small human share is not a general substitute for full human labelling. A natural idea is to send the human labels to the sentences the student is least sure about. At 25% this does not help: routed ground truth scores 59.49 against 64.87 for a random share on BC5CDR, 62.86 against 65.18 on CleanCoNLL, and 75.70 against 75.03 on MIT Movie.

Q7: cost and speed. Figure 3 places students and teacher on one chart for NER, pooled over

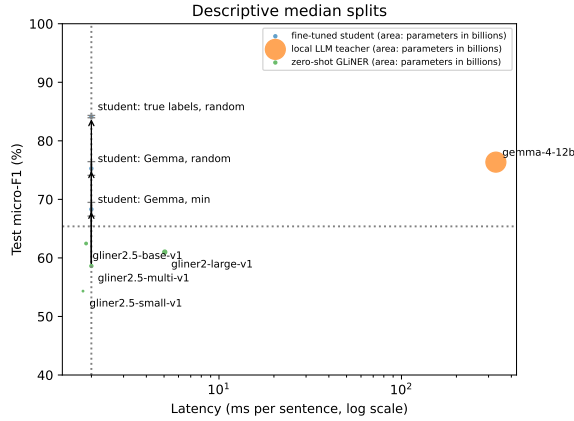


Figure 3: NER, pooled over the three datasets: test micro-F1 against latency on one RTX 3090. Dotted lines are medians. Arrows lead from the zero-shot student to fine-tuned students (400 sentences). Error bars: seed SD.

the three datasets. Gemma 4 12B labels at 329.04 ms per sentence and reaches 76.36 F1. A student trained on 400 random Gemma-labelled sentences reaches 75.28 at 2.00 ms, 165 times faster; the estimated inference cost falls from 20.11 to 0.12 USD per million sentences. The same student trained on 400 ground-truth sentences reaches 84.14, and on 400 Gemma labels picked by min only 68.31; the untrained student scores 58.62. The Pareto front (Figure 4), learning curves (Figure 5) and per-dataset cost curves, which add labelling and selection cost (Figure 6), are in Appendix A.

4.1 Replicating the thesis experiments

We reran the thesis experiments on all seven dataset cells with the protocol above: three seeds, a frozen pool, and dev for every choice.

Thesis selectors measure noise. The thesis compared three sentence scores: mean confidence (avg), mean squared error from full confidence (MSE) and mean negative log-probability (MNLP). In all 63 cells (dataset, label source, budget and seed) the three scores select exactly the same sentences (63 of 63). Each score ranks a sentence with no prediction as least confident, and every pool holds more such sentences than the largest thesis budget: from 563 (6.40%) on MIT Movie to 10,545 (87.01%) on Hallmarks. The selected set is therefore a seeded random draw of empty sentences. For NER, min uses the same rule and selects the same set too (27 of 27 cells). Any score difference between avg, MSE and MNLP is training noise, which makes them a free repeatabil-

ity test: with the same sentences and seeds, mean F1 differs by up to 4.02 points on Hallmarks, 1.59 on CrossRE and 1.46 on MASSIVE English, but only 0.00 on BC5CDR with ground truth. The thesis ranking of these scores cannot be read as a ranking of methods.

Which weights to train. The default student trains LoRA adapters on the encoder and the task heads. With 400 random ground-truth sentences (three seeds), adapters on the encoder alone score within one point of the default on every dataset (MIT Movie 84.08 against 84.23, Hallmarks 43.67 against 42.98). Training the heads alone loses heavily everywhere (CleanCoNLL 65.14 against 85.66, MASSIVE English 50.07 against 69.98, BC5CDR 73.37 against 82.53) and collapses on Hallmarks (4.13). The encoder carries the adaptation. Full fine-tuning of all weights on the whole ground-truth pool (one seed, learning rate not tuned) scores below LoRA on the whole pool for BC5CDR (85.28 against 87.36), CleanCoNLL (88.79 against 89.63), MIT Movie (87.24 against 88.36) and MASSIVE English (71.88 against 75.36). It is higher only on Hallmarks (51.41 against 42.32), where the LoRA run stopped early (Limitations). LoRA is enough for this student.

Thresholds and budgets. The thesis tuned the decision threshold on test data. On dev, the threshold matters for the zero-shot student and hardly at all after training: dev F1 moves by up to 11.92 points across thresholds 0.3 to 0.7 for the zero-shot student on Hallmarks (11.18 on MASSIVE English), against at most 1.99 for a trained student on any dataset. The learning curves over budgets from 50 to 1,000 sentences are in Figure 5.

5 What the thesis did, and what this study fixes

The thesis ran sixteen experiments on MIT Movie: zero-shot and fully fine-tuned GLiNER, an Optuna search and a LoRA layer study, confidence brackets and a threshold sweep, four selection scores over fourteen budgets, a hard-example comparison of GLiNER and Gemma 3, students on LLM versus ground-truth labels, a mixing grid of ground truth and LLM labels, four teachers, synthetic data, a prompt validator and a cost study. It found, without stating it openly, that Gemma 3’s F1 drops from 69.4 on the full test set to 56.9 on the 100 training sentences where GLiNER was least con-

fident; this study turns that observation into question Q2.

The thesis protocol had problems that this study fixes: the MIT Movie dev file was a copy of the test file, so Optuna, early stopping and threshold choice used test data; the pool filter removed sentences with no ground-truth entity; the saved adapter was the last checkpoint rather than the best; the Optuna pruner never received intermediate scores; and no result was repeated with a second seed. Rerun under the fixed protocol on all datasets (Section 4.1), two thesis conclusions do not hold: its three selection scores pick the same sentences, so their ranking was noise, and its mixing result holds on MIT Movie only.

6 Limitations

Our results cover one round of selection, one student family and English plus French data. CrossRE uses given entity mentions, so its results do not cover end-to-end relation extraction. The hyphen-free tokenisation of the datasets hides a GLiNER2 word-splitting issue that matters for raw biomedical text. We cannot rule out that our models saw the evaluation data. No model owner states that our six datasets were in training, and none states that they were removed. The GLiNER2 family reports zero-shot results on CrossNER, whose sentences and splits CrossRE reuses, so CrossRE test sentences were part of the student developers’ entity evaluation; whether checkpoints were chosen on them is unknown. The student’s training text includes news and PubMed, so the text (not the labels) of CleanCoNLL, BC5CDR and Hallmarks may overlap. CoNLL-2003 is a known contamination case for earlier LLMs, and public instruction collections contain CoNLL and BC5CDR tasks; our teachers do not document whether such data was used. MASSIVE postdates the student encoder’s pre-training text. The full audit is in Appendix B. Training is not exactly repeatable on our GPU. NER runs on ground truth repeat almost to the digit (identical selections differ by at most 0.01 points), but relation, classification and slot runs, and NER runs on Gemma labels, can differ by up to 4.02 points on the same sentences and seed (Section 4.1). We therefore report the seed SD of every cell and of every contrast. Costs are estimates: GPU time at one rental price on one date, and ground-truth labels at a simulated crowd rate of 0.50 USD per sentence taken

from a span-annotation study (Kasner et al., 2026). The planned API teachers (Qwen and gpt-oss on Cerebras) were not run, so every teacher in this study is a local Gemma model, and the smaller E4B teacher labelled only the ladder samples and two training cells.

Three further points qualify single cells. The whole-pool Hallmarks run stopped early on a dev plateau, so it understates what the full pool can give; sentence-level Hallmarks is also much harder than the abstract-level task on which published scores are reported. Six CrossRE runs stopped with a non-finite gradient and one MASSIVE run made no optimiser step; all ran again with the same configuration and seed and finished normally. Runs killed by a GPU fault or out of memory were restarted in a new process with the same configuration and seed. The GLiNER2 trainer silently skips a batch that runs out of GPU memory; a check of epoch against step count in every training log found five runs with skipped batches. We made out of memory stop the run, and those five runs were repeated in full. Finally, fine-tuned students were not timed separately: they reuse the zero-shot timing of their base model, on the unmeasured assumption that a LoRA adapter adds little inference time.

Ethics and broader impact

All datasets are public and used for research as intended by their licences. We release teacher labels as sentence ids and character offsets, never text whose licence forbids it. No human annotators were involved. The GPU time of every run is in the released run logs.

Use of AI assistants

The first version of the code was written by an AI coding assistant (OpenAI Codex) and reviewed by a second assistant (Anthropic Claude), which wrote the later code, fixes and analysis, helped plan experiments and drafted text. The author checked the design, the results and every citation, and takes responsibility for the content.

References

Simon Baker, Ilona Silins, Yufan Guo, Imran Ali, Johan Högberg, Ulla Stenius, and Anna Korhonen. 2016. *Automatic semantic classification of scientific literature according to the hallmarks of cancer*. *Bioinformatics*, 32(3):432–440.

- Elisa Bassignana and Barbara Plank. 2022. CrossRE: A Cross-Domain Dataset for Relation Extraction. *arXiv preprint arXiv:2210.09345*.
- Zihan Chen, Lei Shi, Weize Wu, Qiji Zhou, and Yue Zhang. 2025. ALLabel: Three-stage Active Learning for LLM-based Entity Recognition using Demonstration Retrieval. *arXiv preprint arXiv:2509.07512*.
- Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. 2018. *The Hitchhiker’s Guide to Testing Statistical Significance in Natural Language Processing*. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1383–1392.
- Rotem Dror, Segev Shlomov, and Roi Reichart. 2019. *Deep Dominance - How to Properly Compare Deep Neural Models*. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 2773–2785.
- Fastino. 2026. GLiNER2.5 multi v1. <https://huggingface.co/fastino/gliner2.5-multi-v1>. Revision 12fc4039.
- Jack FitzGerald, Christopher Hensch, Charith Peris, Scott Mackie, Kay Rottmann, Ana Sanchez, Aaron Nash, Liam Urbach, Vishesh Kakarala, Richa Singh, Swetha Ranganath, Laurie Crist, Misha Britan, Wouter Leeuwis, Gokhan Tur, and Prem Nataraajan. 2022. MASSIVE: A 1M-Example Multilingual Natural Language Understanding Dataset with 51 Typologically-Diverse Languages. *arXiv preprint arXiv:2204.08582*.
- Galileo. 2022. MIT Movies (fixed), CoNLL format. https://huggingface.co/datasets/rungalileo/mit_movies_fixed_conll_format. Revision bf6c430a.
- Abhishek Ghose and Emma Thuong Nguyen. 2024. On the Fragility of Active Learners for Text Classification. *arXiv preprint arXiv:2403.15744*.
- Jonas Golde, Patrick Haller, and Alan Akbik. 2025. FiNERweb: Datasets and Artifacts for Scalable Multilingual Named Entity Recognition. *arXiv preprint arXiv:2512.13884*.
- Ahmad Dawar Hakimi, Lea Hirlimann, Isabelle Augenstein, and Hinrich Schütze. 2026. Do We Still Need Humans in the Loop? Human vs. LLM Annotation in Active Learning for TikTok Hate Speech Detection. *arXiv preprint arXiv:2604.13899*.
- Wataru Hashimoto, Hidetaka Kamigaito, and Taro Watanabe. 2024. Are Data Augmentation Methods in Named Entity Recognition Applicable for Uncertainty Estimation? *arXiv preprint arXiv:2407.02062*.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. LoRA: Low-Rank Adaptation of Large Language Models. *arXiv preprint arXiv:2106.09685*.
- Zdeněk Kasner, Vilém Zouhar, Patrícia Schmidtová, Ivan Kartáč, Kristýna Onderková, Ondřej Plátek, Dimitra Gkatzia, Saad Mahamood, Ondřej Dušek, and Simone Balloccu. 2026. LLMs as span annotators: A comparative study of LLMs and humans. In *Proceedings of the MME Workshop at EACL 2026*. ArXiv:2504.08697.
- Nataliia Kholodna, Sahib Julka, Mohammad Khodadadi, Muhammed Nurullah Gumus, and Michael Granitzer. 2024. LLMs in the Loop: Leveraging Large Language Model Annotations for Active Learning in Low-Resource Languages. *arXiv preprint arXiv:2404.02261*.
- Philipp Kohl, Yoka Krämer, Claudia Fohry, and Bodo Kraft. 2024. Scoping Review of Active Learning Strategies and their Evaluation Environments for Entity Recognition Tasks. *arXiv preprint arXiv:2407.03895*.
- Jiao Li, Yueping Sun, Robin J. Johnson, Daniela Sciaky, Chih-Hsuan Wei, Robert Leaman, Allan Peter Davis, Carolyn J. Mattingly, Thomas C. Wieggers, and Zhiyong Lu. 2016. *BioCreative V CDR task corpus: a resource for chemical disease relation extraction*. Database, 2016:baw068.
- Jingjing Liu, Panupong Pasupat, Scott Cyphers, and James Glass. 2013. *Asgard: A portable architecture for multilingual dialogue systems*. In *Proceedings of the 2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 8386–8390.
- Viviana Luccioli, Rithika Iyengar, Ryan Panley, Flora Haberkorn, Xiaoyu Ge, Leland Crane, Nitish Sinha, and Seung Jung Lee. 2025. LLM on a Budget: Active Knowledge Distillation for Efficient Classification of Large Text Corpora. *arXiv preprint arXiv:2511.11574*.
- Elena Merdjanovska, Jonas Golde, and Alan Akbik. 2026. Error-Type-Aware Loss Reweighting for Robust Named Entity Recognition with Noisy LLM Labels. *arXiv preprint arXiv:2608.30827*.
- Sudhanshu Mittal, Maxim Tatarchenko, Özgün Çiçek, and Thomas Brox. 2019. Parting with Illusions about Deep Active Learning. *arXiv preprint arXiv:1912.05361*.
- Yuanyuan Qi, Xiaohao Yang, Jueqing Lu, Guoxiang Guo, Joanne Enticott, Gang Liu, and Lan Du. 2026. Next Generation Active Learning: Mixture of LLMs in the Loop. *arXiv preprint arXiv:2601.15773*.
- Julia Romberg, Christopher Schröder, Julius Gonsior, Katrin Tomanek, and Fredrik Olsson. 2025. Re-assessing Active Learning Adoption in Contemporary NLP: A Community Survey. *arXiv preprint arXiv:2503.09701*.

Susanna Rücker and Alan Akbik. 2023. CleanCoNLL: A Nearly Noise-Free Named Entity Recognition Dataset. *arXiv preprint arXiv:2310.16225*.

Yanyao Shen, Hyokun Yun, Zachary C. Lipton, Yakov Kronrod, and Animashree Anandkumar. 2017. Deep Active Learning for Named Entity Recognition. *arXiv preprint arXiv:1707.05928*.

Abdelkrim Tafer, Mauro Gaio, and Javier Lacasta. 2026. A Hybrid Annotation Method and a Reference Corpus for Hiking Descriptions. In *Proceedings of the Fourth International Workshop on Geographic Information Extraction from Texts (GeoExT 2026)*, volume 4201 of *CEUR Workshop Proceedings*, pages 89–98, Delft, The Netherlands.

Erik F. Tjong Kim Sang and Fien De Meulder. 2003. Introduction to the CoNLL-2003 Shared Task: Language-Independent Named Entity Recognition. In *Proceedings of the Seventh Conference on Natural Language Learning at HLT-NAACL 2003*, pages 142–147.

Dennis Ulmer, Christian Hardmeier, and Jes Frellsen. 2022. deep-significance - Easy and Meaningful Statistical Significance Testing in the Age of Neural Networks. *arXiv preprint arXiv:2204.06815*.

Karthik S. Vedula, Annika Gupta, Akshay Swaminathan, Ivan Lopez, Suhana Bedi, and Nigam H. Shah. 2024. Distilling Large Language Models for Efficient Clinical Information Extraction. *arXiv preprint arXiv:2501.00031*.

Anthony Yazdani, Ihor Stepanov, and Douglas Teodoro. 2025. GLiNER-BioMed: A Suite of Efficient Models for Open Biomedical Named Entity Recognition. *arXiv preprint arXiv:2504.00676*.

Simin Yu and Sufia Fathima. 2026. When Active Learning Falls Short: An Empirical Study on Chemical Reaction Extraction. *arXiv preprint arXiv:2604.19335*.

Urchade Zaratiana, Gil Pasternak, Oliver Boyd, George Hurn-Maloney, and Ash Lewis. 2025. GLiNER2: An Efficient Multi-Task Information Extraction System with Schema-Driven Interface. *arXiv preprint arXiv:2507.18546*.

Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. 2023. GLiNER: Generalist Model for Named Entity Recognition using Bidirectional Transformer. *arXiv preprint arXiv:2311.08526*.

Hao Zeng, Huipeng Huang, Xinhao Qu, Jianguo Huang, Bingyi Jing, and Hongxin Wei. 2026. HyPAC: Cost-Efficient LLMs-Human Hybrid Annotation with PAC Error Guarantees. *arXiv preprint arXiv:2602.02550*.

Ruoyu Zhang, Yanzeng Li, Yongliang Ma, Ming Zhou, and Lei Zou. 2023. LLMaAA: Making Large Language Models as Active Annotators. *arXiv preprint arXiv:2310.19596*.

Zihao Zheng, Baichuan Li, Junyi Yao, and Jiayu Long. 2026. Reliable Financial Named Entity Recognition Under Domain Shift: Confidence Estimation and Selective Prediction. *arXiv preprint arXiv:2608.19558*.

A Additional figures

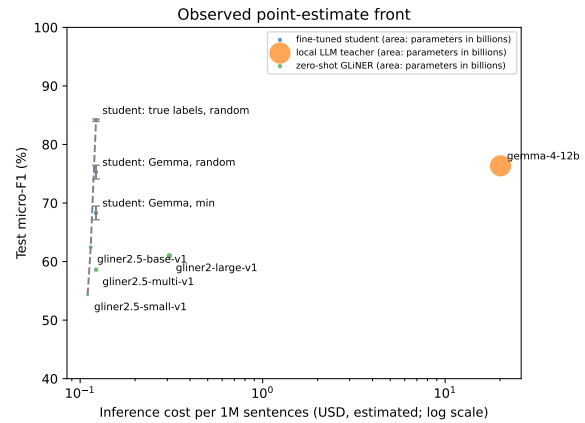


Figure 4: NER, pooled over three datasets: test micro-F1 against estimated inference cost per million sentences (GPU time at one rental price). The dashed line joins the observed points that no other point beats on both axes.

B Data overlap audit

We checked papers, model cards and technical reports (September 2026) for each pair of evaluation dataset and model. We ran no probe, so nothing in this section is measured. Table 4 gives the result. No model owner states that any of our datasets was in training, and none states that it was removed.

Dataset	Student	Encoder	Teachers
CleanCoNLL	U	U	P, high
BC5CDR	U	P	P
MIT Movie	U	U	P, low
CrossRE	E	P	P, high
Hallmarks	U	P	P
MASSIVE	U	N	P

Table 4: Possible overlap of evaluation data with model training data. E: evaluated on by the model owner; P: a named path exists but no source confirms it; N: not possible by date; U: unknown. Student: GLiNER2.5-multi; encoder: mDeBERTa-v3 (CC-100); teachers: Gemma 4 E4B and 12B.

Student. The GLiNER2.5 release states that its training data was generated by an agent rather than assembled from existing datasets; it names no

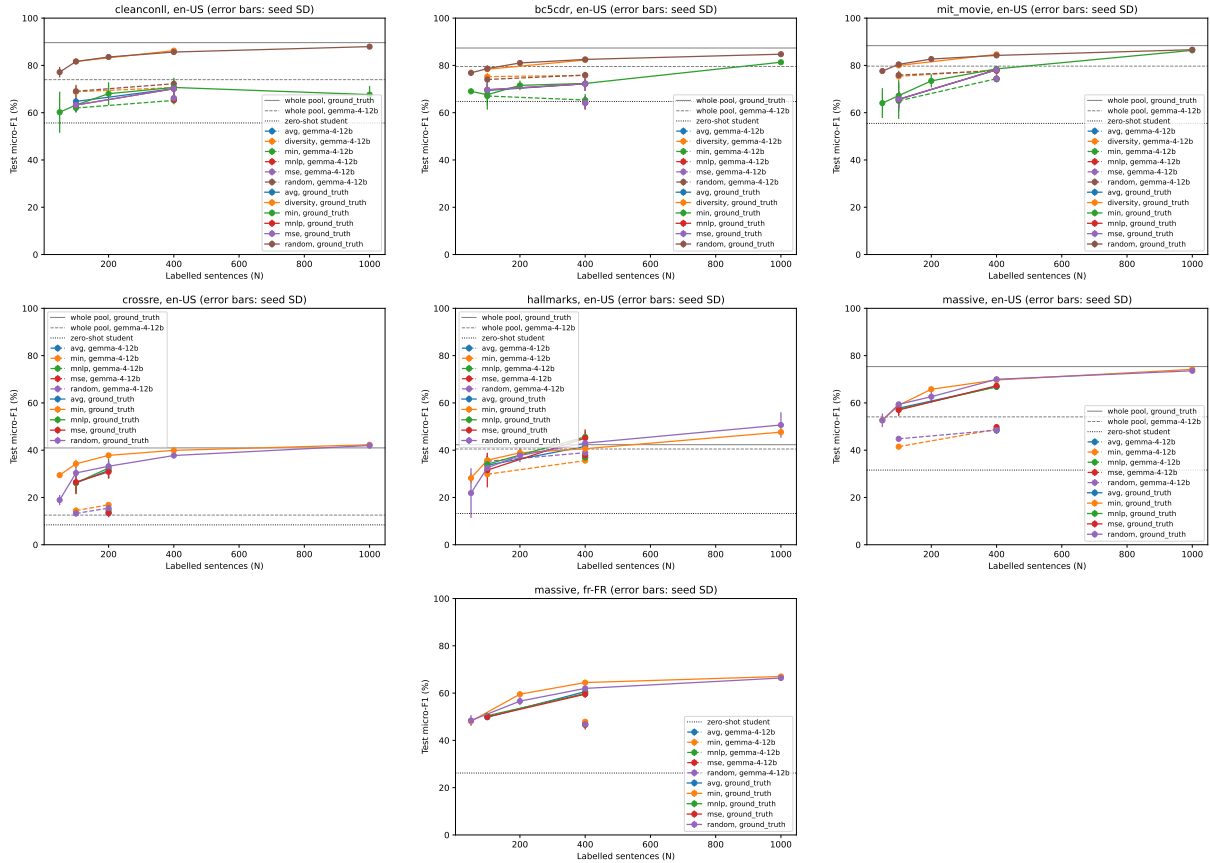


Figure 5: Test micro-F1 against the number of labelled sentences. Solid lines: ground truth; dashed lines: Gemma 4 12B labels. Grey lines: whole pool; dotted line: zero-shot student. Error bars: seed SD.

teacher, corpus or size and makes no decontamination statement (Fastino, 2026). It reports zero-shot results on five CrossNER domains. CrossRE annotates relations on the CrossNER sentences with the same splits (Bassignana and Plank, 2022), so the student’s developers have scored entities on our CrossRE test sentences; whether they chose checkpoints on those scores is unknown. GLiNER2 trains on news, Wikipedia and PubMed text labelled by GPT-4o (Zaratiana et al., 2025), so the text, not the labels, of BC5CDR, Hallmarks and CrossRE may overlap with the zero-shot ladder models.

Encoder. mDeBERTa-v3 is pre-trained on CC-100, built from 2018 Common Crawl snapshots. MASSIVE (2022) and its source SLURP (2020) cannot be in it. Wikipedia and PubMed pages can be, so CrossRE, BC5CDR and Hallmarks text may be familiar to the encoder; a masked encoder sees text without our labels.

Teachers. Every dataset was public before the Gemma 4 data cutoff (January 2025). The Gemma 4 report states that benchmarks were de-

contaminated but does not list them. CoNLL-2003 is a documented contamination case for ChatGPT, which reproduces its test sentences with labels. Public instruction collections contain CoNLL and BC5CDR tasks; no teacher owner says whether it used them. We therefore treat CleanCoNLL and the news domain of CrossRE, which is CoNLL-2003 text, as likely seen by the teachers.

Effect on our claims. Exposure raises absolute teacher and student scores. Our main claims compare selection arms, and teachers, on the same sentences. Exposure that is equal across arms does not bias these paired contrasts, but it can narrow the measured gap between teacher labels and ground truth. MASSIVE English and French are translations of the same utterances, so the two locales are not independent evidence.

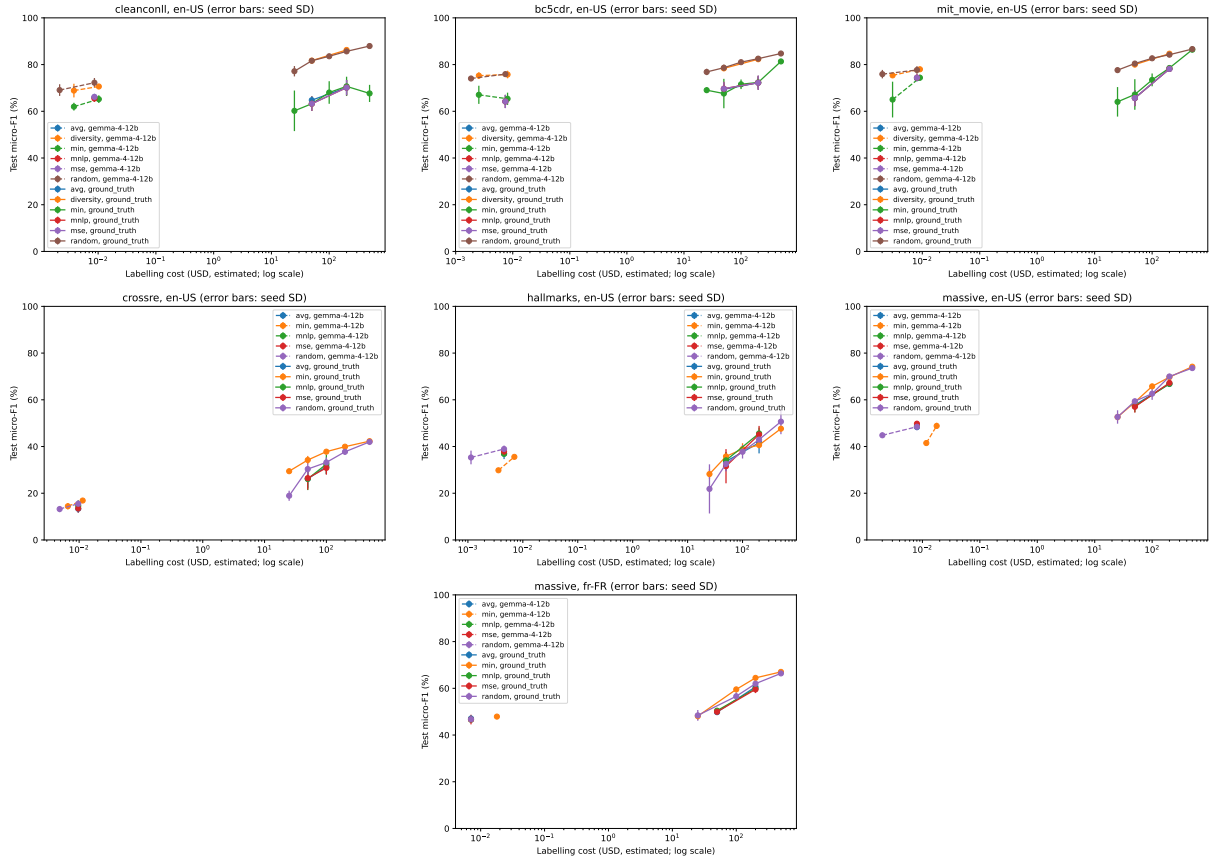


Figure 6: Test micro-F1 against estimated labelling cost: ground truth at a simulated 0.50 USD per sentence, Gemma labels at free-GPU time, plus pool scoring for min and diversity.